
JNML.CO
Red, Blue, Purple Teaming Exercise Report

TABLE OF CONTENTS **Pg No.**

Introduction **3**
Team Members 3
Workflow 3

RED TEAM Report
1.0 Methodology of approach 4
2.0 Scope and planning 4
2.1 Scenario 4
2.2 Scope 4
2.3 Plan of attack 4
3.0 Attack narrative 4
3.1 Scan the network using Nmap 5
3.2 Open ports 6
3.3 Enumeration 6
3.4 Decode “README.md” file 8
3.5 Exploitation 10
4.0 Access obtained and Data exfiltrated 12
4.1 MITRE ATT&CK & Tactics, Techniques & Procedures (TTPs) 13
5.0 Observations and recommendations 13
5.1 Observations 13
5.2 Recommendations 13

BLUE TEAM Incidence Response Plan
1.0 Response 14
2.0 Incident Response Template 16
3.0 System identifications 17
4.0 Impact of security incident 17
5.0 Actions taken for containment, recovery, and restoration of services. 18
6.0 Communication with stakeholders and relevant authorities 18
7.0 Lessons learned and recommendations for future incidents. 19

PURPLE TEAM Analysis
1.0 Introduction 20
2.0 Background 20
3.0 Timeline 21
4.0 Red Team review 22
5.0 Blue Team review 22
6.0 Purple Team exploits automation (optional) 23
7.0 Purple Team actions taken. 24
8.0 Recommended presentation 25
9.0 Conclusion 25

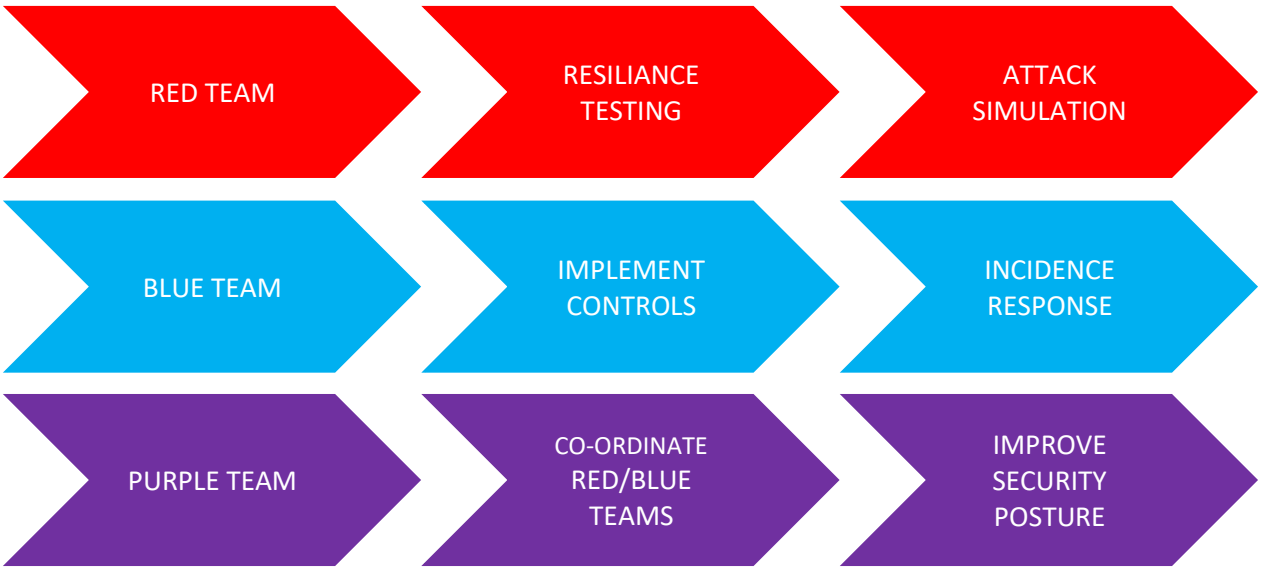
Introduction:

This report details the results of the Red, Blue, and Purple Teaming exercise conducted by **JNML.CO**. The purpose of the exercise was to simulate a cyber-attack on ABC Retail store and assess the company's ability to respond in order to improve its security posture.

Team Members:

Red Team Members:	Blue Team Members:	Purple Team Members:
1. [redacted] 2. Or	1. [redacted] 2. [redacted]	1. [redacted] 2. [redacted]

Workflow:





RED TEAM Report

1.0 Methodology of approach

The Red Team of JNML.CO conducted an investigation on the system of ABC Retail Store on 2 September 2023.

The Red Team plays a pivotal role in mimicking the approach of real hackers to assess the security of the system. The primary roles can be categorised into three key areas. Firstly, they focus on identifying and analysing security vulnerabilities within systems or network. Secondly, they act as OSINT collectors, responsible for gathering information and data from publicly available sources to gain insights onto the target system. Lastly, they devise and executes various attack scenarios to evaluate the system's security and explore methods of penetration.

The Red Team engagement utilised tactics to target the systems being assessed. This methodology encompassed a series of steps, starting with gathering open-source intelligence (OSINT), followed by system enumeration, exploitation, and executing attacks to achieve specific operational objectives. The objectives of the Red Team assessment were to discover and record any weaknesses that could potentially lead to manipulation or mistakes resulting in data breaches. Additionally, it aimed to obtain user credential details that might allow for the modification and leakage of sensitive data, such as credit card details, user credentials, and other payment methods.

2.0 Scope and planning

2.1 Scenario:

The Red Team engagement was based on the Assumed Breach Model employing port scanning tool, Nmap, to discover the details and vulnerabilities of network.

Nmap can discover to identify active hosts, port scanning to reveal open ports and services, and operation system detection. Additionally, Nmap supports scripting for custom analysis, aids in visualising network structures.

2.2 Scope:

The scope identified by ABC Retail Store encompasses any domain, IP addresses, and subnet that is registered to ABC Retail Store. The target Domain Name is “abc.com”.

2.3 Plan of attack:

For this engagement, the following were the attack plan:

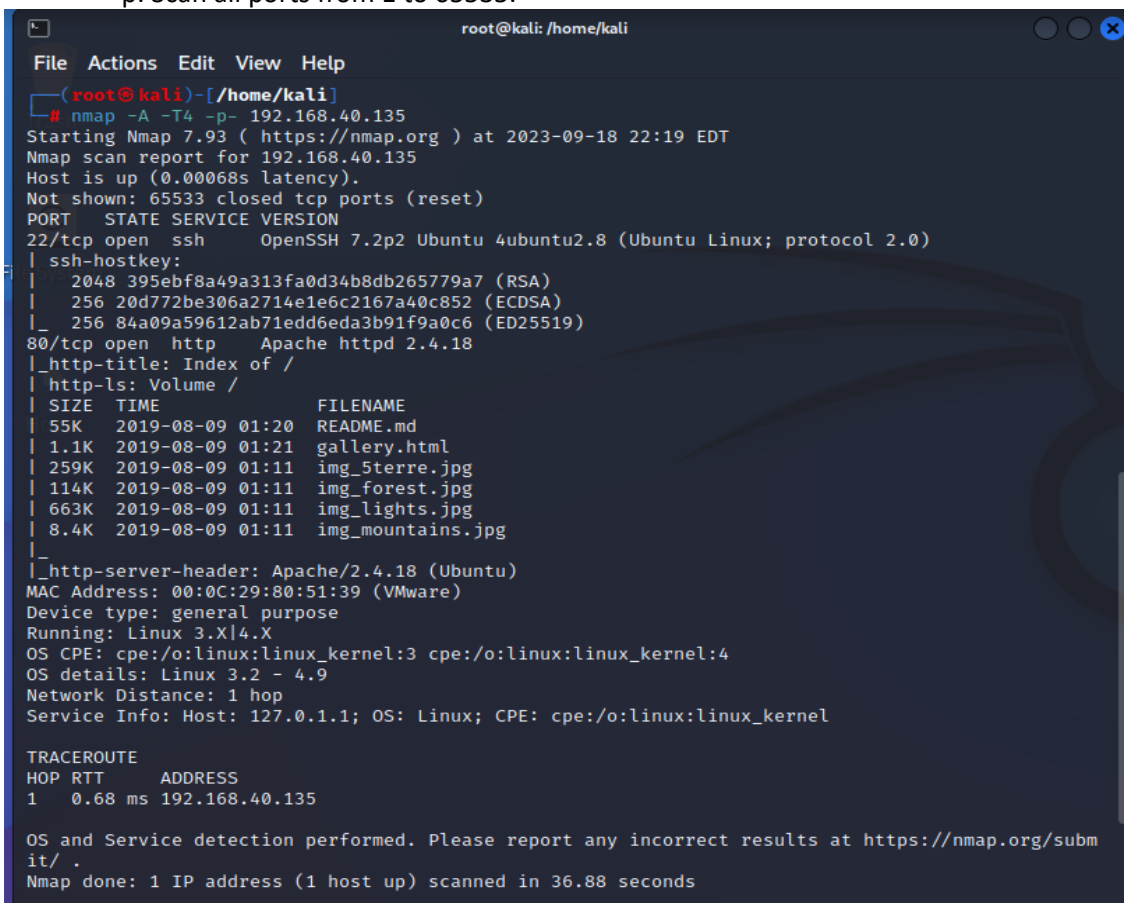
- Extensively delve into OSINT to gather comprehensive data regarding abc.com.
- Explore all servers registered to the abc.com domain to pinpoint a potential security weakness that can be exploited to gain system access.
- Look for all the subnets, ports, services, and Ips linked to abc.com.

3.0 Attack narrative

3.1 Scan the network using Nmap:

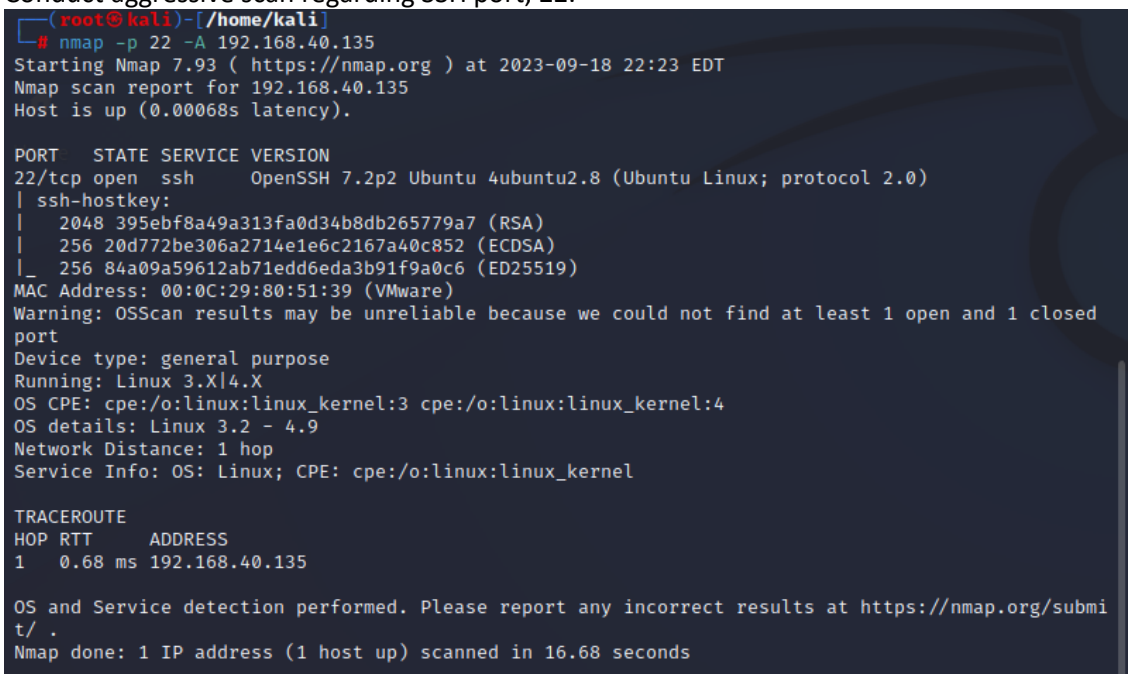
nmap -A -T4 -p- 192.168.40.135

- A: Aggressive scan. It includes operating system and service version detection, script scanning and runs traceroute for more detailed information gathering.
- T4: Scan speed setting. It represents a faster scan.
- p: Scan all ports from 1 to 65535.



nmap -p 22 -A 192.168.40.135

Conduct aggressive scan regarding SSH port, 22.



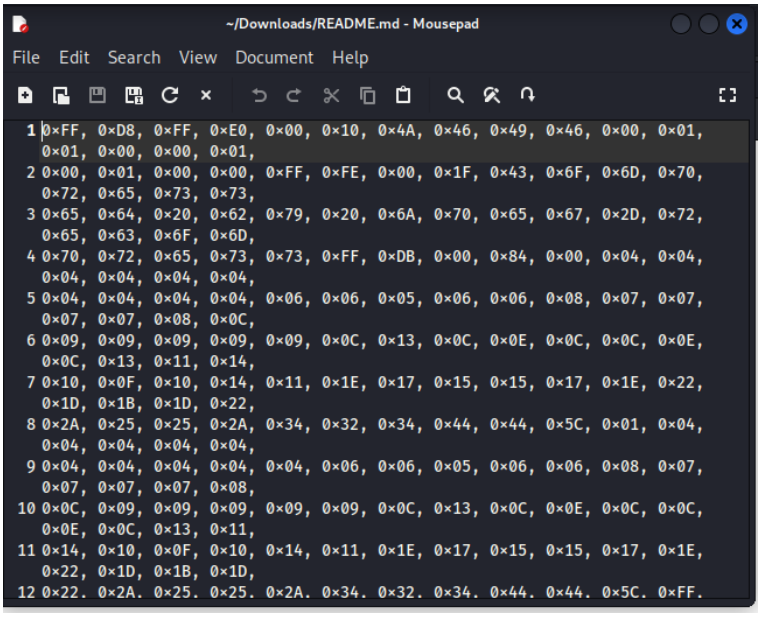
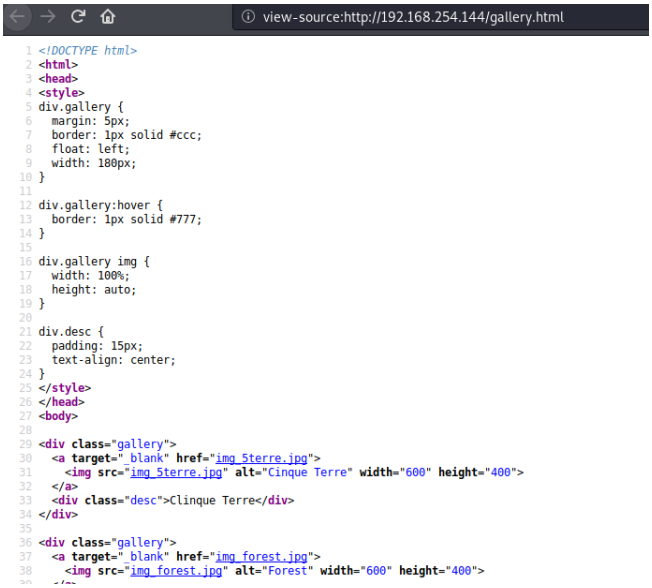
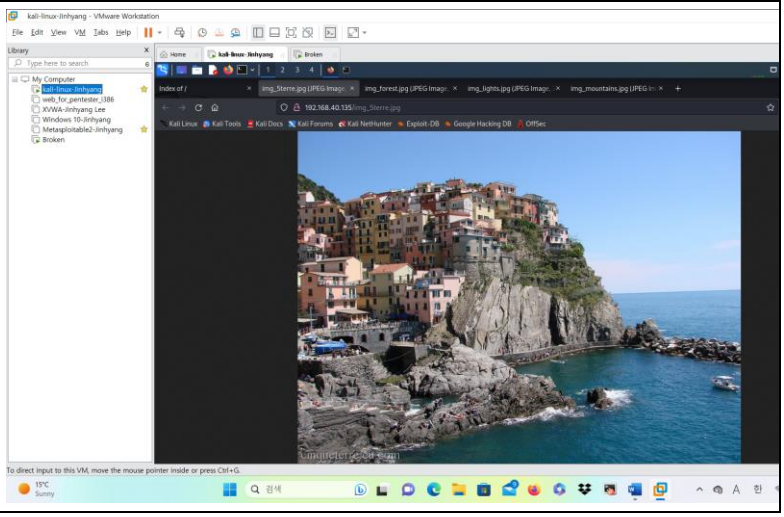
3.2 Open ports:

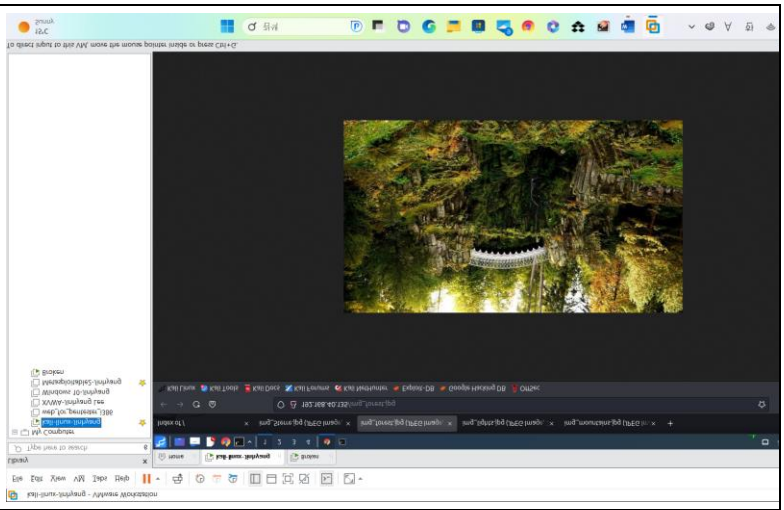
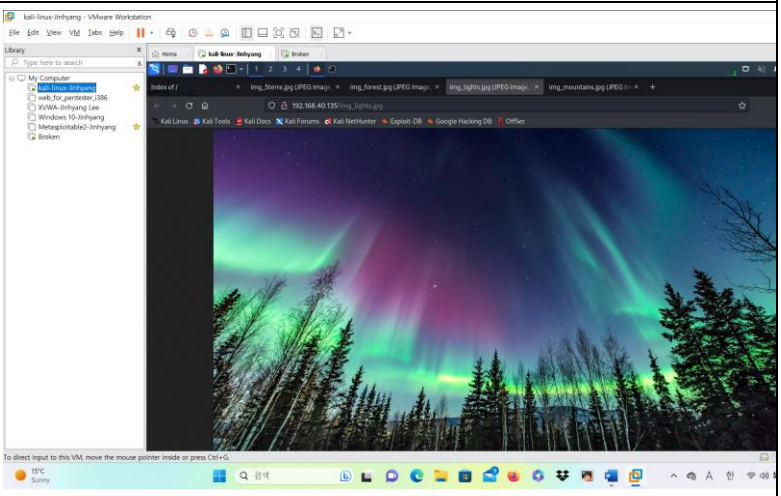
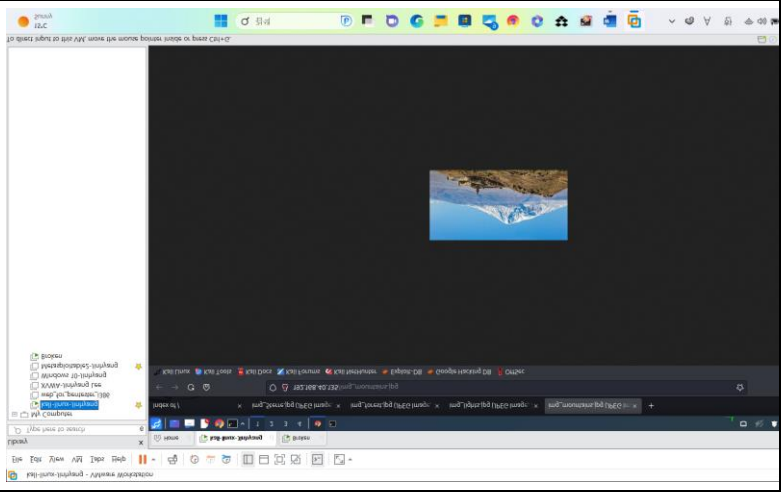
	TCP or UDP	Port number	
SSH	TCP	Port 22	Secure Shell is a well-known protocol which provides a secure way to access and manage remote computers over the Internet
HTTP	TCP	Port 80	Hypertext Transfer Protocol is a protocol for data communication on the World Wide Web, enabling the transfer of web pages and resources from web servers to web browsers for viewing and interaction

3.3 Enumeration:

Through direct access to the target website using the HTTP protocol, the Red Team successfully conducted reconnaissance on the system. We accessed the website hosted at the IP address 192.168.40.135, where we discovered a total of six files. Below, the detailed information about these files are displayed, including their names, types, and accompanying screenshots.

#	File name	Type of file (text, image)	Screenshot of the file content

1	README.md	Text file	
2	Gallery.html	Text file with html code	
3	Img_5terre.jpg	image	

4	Img_forest.jpg	Image	
5	Img_lights.jpg	Image	
6	Img_mountains.jpg	image	

3.4 Decode “README.md” file

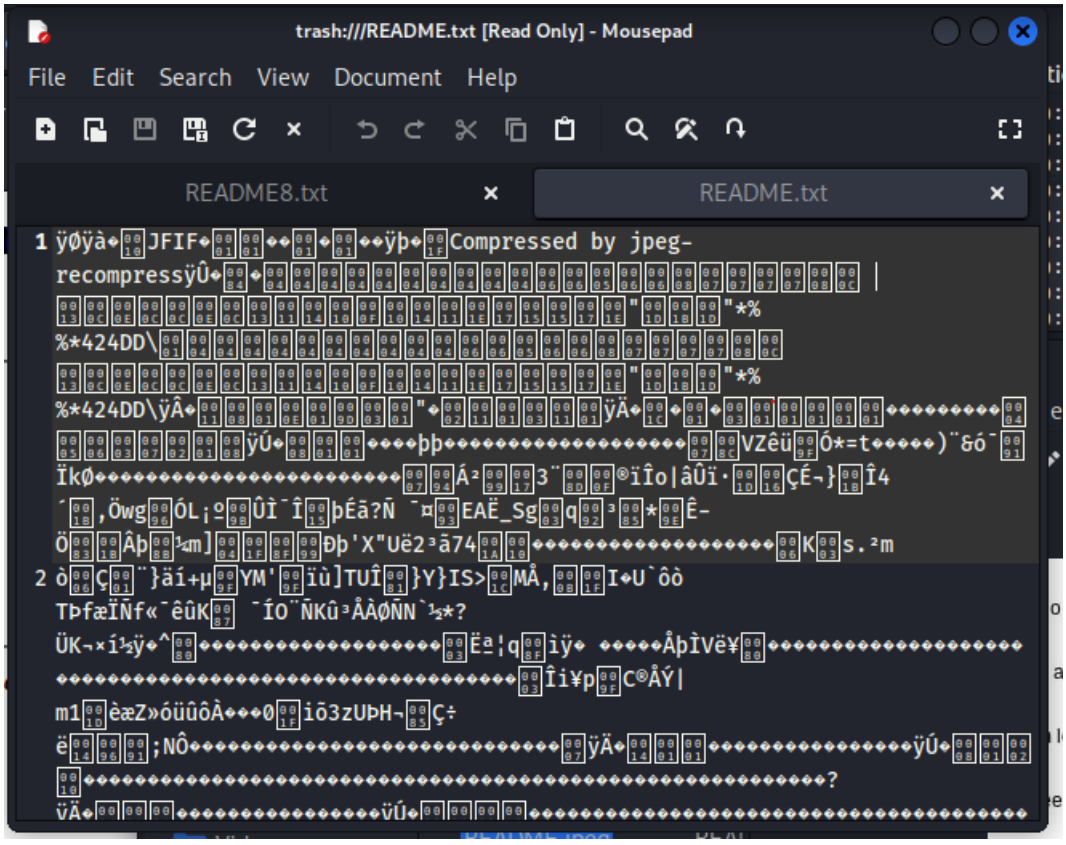
During reconnaissance, the Red Team encountered files with clear and recognizable content, such as .html and .jpg files. However, the file “README.md” is mysterious. Upon examination, we found that this file contained a series of hexadecimal numbers. This led to the theory that each octet in this file represented an ASCII character. To validate this hypothesis, we convert the hexadecimal numbers into their corresponding ASCII characters using kali linux. This process aimed to unveil the hidden message within the “README.md”file.

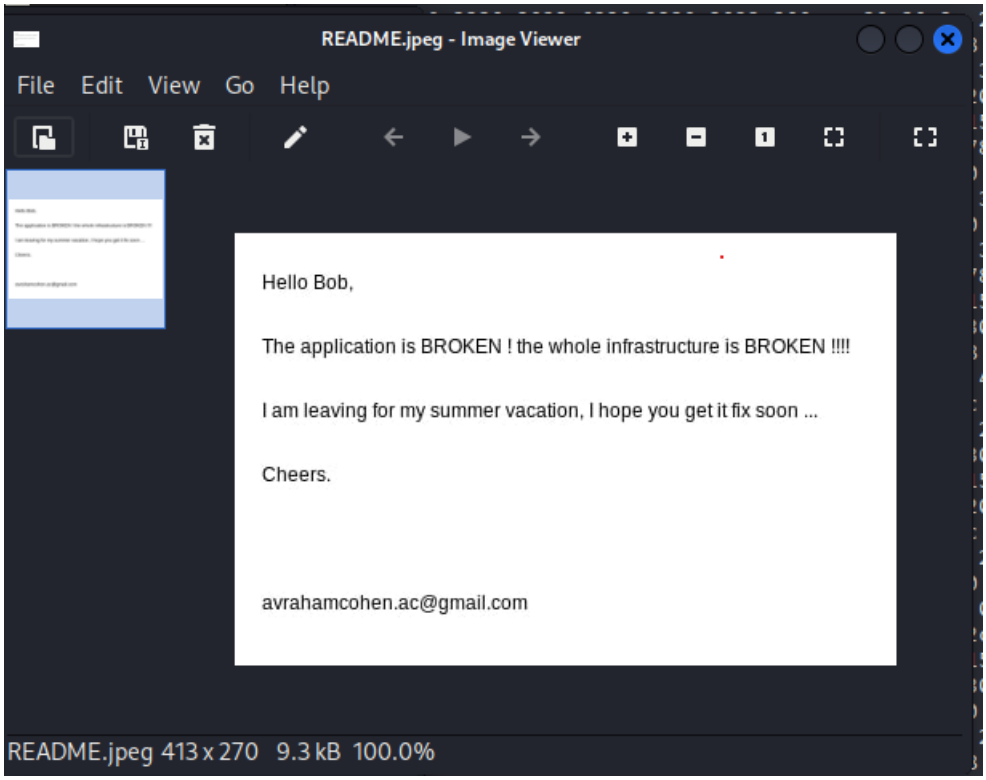
xxd -r -p README.md > README.txt

xxd	This is the main command, which is used for various tasks related to hexadecimal encoding and decoding.
-r	This option is used to reverse the operation. Convert the hexadecimal data back into binary or text.
-p	This option specifies that the input is in plain hexadecimal format. It's used to indicate that the hexadecimal data in the input file should be processed.
README.md	This is the input file. It's the file that contains the hexadecimal data that we want to convert back to its original form.
> README.txt	This part of the command is used for output redirection. It takes the result of the 'xxd' command and write it to a new file called "README.txt."

```
┌──(root㉿kali)-[/home/kali/Downloads]
└─# xxd -r -p README.md > README.txt
```

Upon opening the file "README.txt", the Red Team made an intriguing discovery. The text within the file revealed a clear message: "Compressed by jpeg". This finding prompted us to convert this data into a .jpeg file.





3.5 Exploitation:

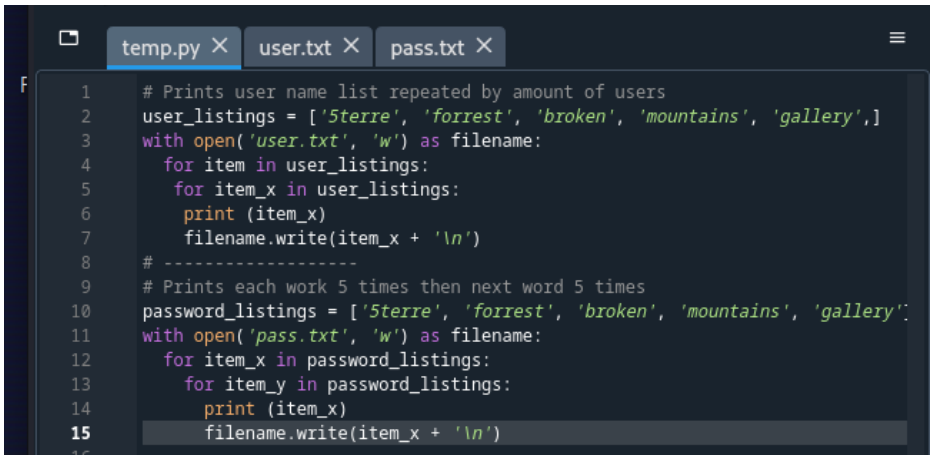
Based on the previous finding, the Red Team identified some key words, such as file name of the pictures, special word in the README.md picture. We prepared a wordlists that would be used in a brute force password attack against the target domain using “Hydra”.
Hydra is a powerful password-cracking tool used in the field of cybersecurity. It is designed for the purpose of performing brute-force and dictionary attacks to gain unauthorised access to the systems, services, and accounts. It supports a wide range of protocols, such as SSH, FTP, HTTP, Telnet, SMB, RDP, and more.

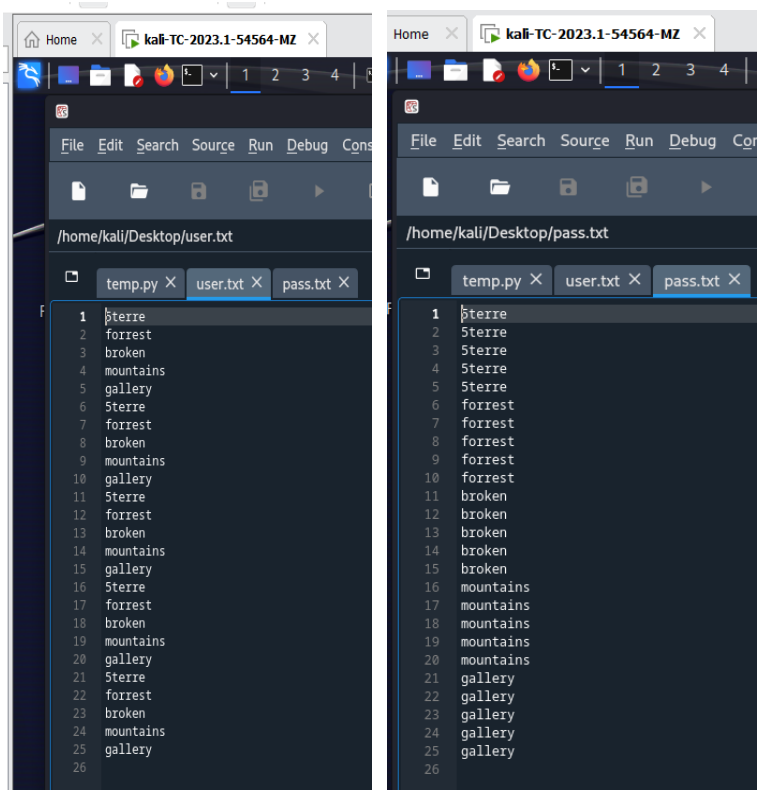
- Create two dictionaries for user ID and password. These will be used in a brute force password attack.

```
┌──(root㉿kali)-[/home/kali/Downloads]
└─# cat > user.txt
5terre
forrest
lights
mountains
broken
gallery
^C
```

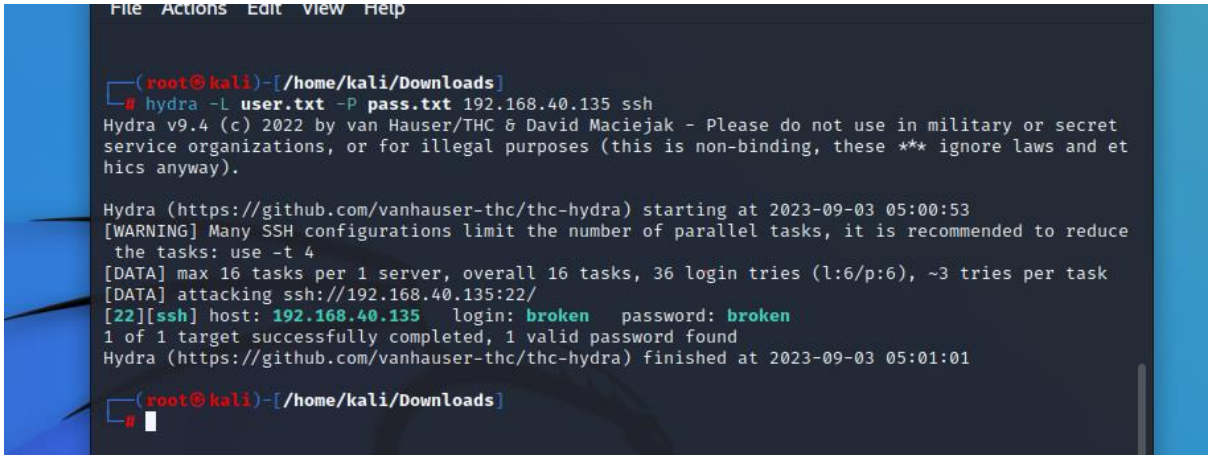
```
┌──(root㉿kali)-[/home/kali/Downloads]
└─# cat > pass.txt
5terre
forrest
lights
mountains
broken
gallery
^C
```

- Programmingly generate username and password file for hydra tool (Exploit automation) When generating the username and password files, we had an assumption that the username and password are the same word, which is not a good enough coverage in terms of the key words combination. The below is a python script that automates password files for hydra tool within the use of The Red Team attack. This assists in accelerating the testing process for future assessments. This shows steps for automation:

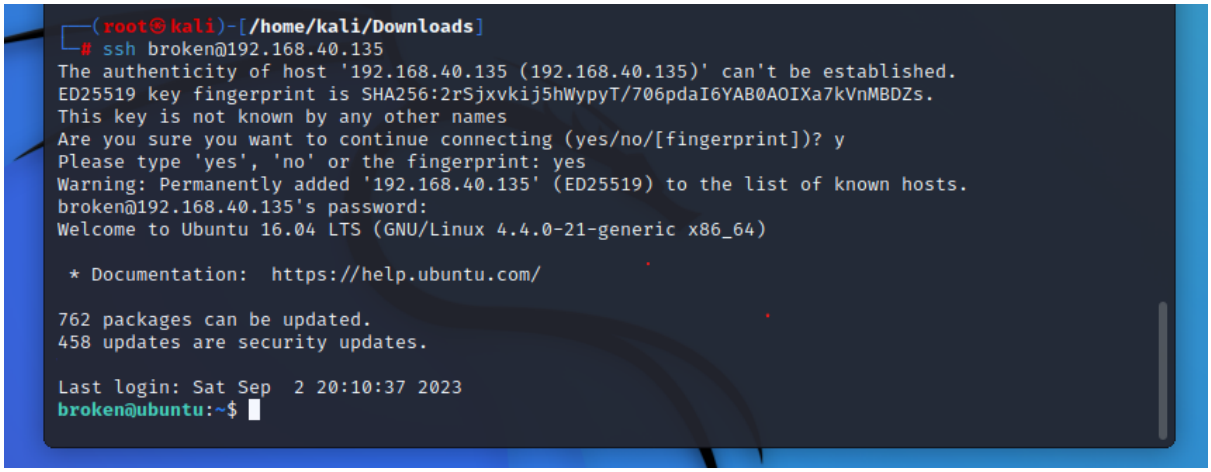




- Launch brute-force password attack.
  Hydra -L user.txt -P pass.txt 192.168.40.135 ssh



- Login using the credentials obtained by brute-force password attack.
  ssh broken@192.168.40.135
  ID: broken, Password: broken



4.0 Access obtained and Data exfiltrated

The following are the list of potential files that the Red Team could have exfiltrate from the network if the Blue Team were not able to defend:

- Sensitive internal employee and customer credentials, such as Email, VPN, SSH, and financial data.
- Critical business-operating and financial information including transactions, third-party orders, and vendors.
- Admin account access that can perform read, write, and execute to all the files in the system, including backup.
- Blue team reports including regular monitoring logs and the countermeasures that have been conducted.

4.1 MITRE ATT&CK & Tactics, Techniques & Procedures (TTPs):

For this investigation, following are the TTPs that were used:

- Reconnaissance – Active scanning TTP ID:T1595
- Resource Development – Network Service Discovery TTP ID: T1046
- Initial Access – External Remote Services TTP ID : T1133
- Credential Access - Brute Force TTP ID : T1110
- Discovery – Account Discovery TTP ID: T1087
- File and Directory Discovery TTP ID : T1083
- Remote System Discovery TTP ID : T1018
- Lateral Movement – Remote Services TTP ID : T 1021.004
- Command and Control – Application Layer Protocol TTP ID:T1071
- Exfiltration – Transfer Data to Cloud Account TTP ID : T1537
- Impact – Data Manipulation TTP ID: T 1565

Reference: https://attack.mitre.org/tactics/enterprise/

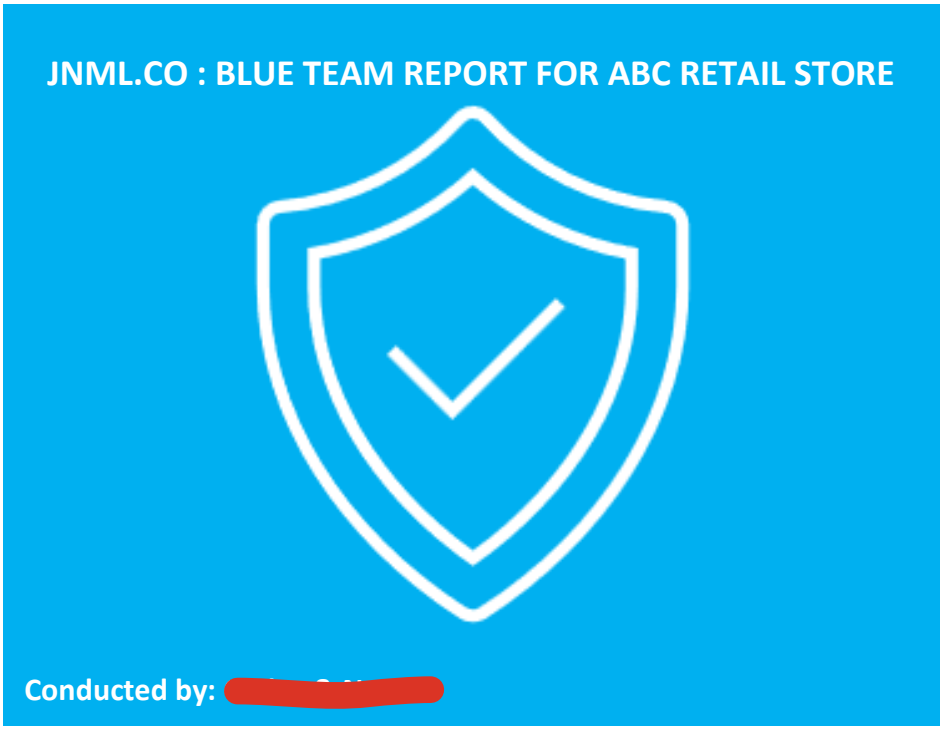
5.0 Observations and recommendations

5.1 Observations:

- The SSH service is running OpenSSH 7.2p2 Ubuntu 4ubuntu2.8.
- The currently used version of SSH is outdated, potentially lacking recent security updates and patches.
- The web server is running Apache httpd 2.4.18 which may lead to data breach depending on configuration and version management.
- Lack of a strong password policy
- Lack of a firewall policy to detect abnormal activities and to protect the system regardless of the presence of an administrator.

5.2 Recommendations:

- Enforce the strong password policy to the entire system to align with the NIST guidelines.
- Implement Multi-Factor Authentication (MFA) and review MFA configurations to guarantee comprehensive coverage for all active, externally accessible protocols.
- Log all the access and activities to the system.
- Strengthen VPN ACLs to restrict user's access only to the assets they are supposed to.
- Allow VPN connections based on hardware addresses.
- Regularly update the operating system and services to the latest versions and enhance security configurations.



BLUE TEAM Incidence Response Plan

During a simulated Red Team exercise involving the Hydra tool, a deliberate and controlled brute-force attack was conducted on the SSH and HTTP services of the target system. This report outlines the immediate response of the blue team, including the activation of the incident response team, and emphasizes that there was no downtime or disruptions to normal operations during the exercise.

1.0 Response:

The Security monitoring systems, which include the IDS with ET SCAN signatures, promptly detected a potential brute-force attack on the SSH and HTTP services and alerted the Blue Team. Concurrently, Wireshark log analysis identified login attempts exhibiting patterns indicative of potential unauthorised access attempts. Subsequently, the Incident Response Team was activated to coordinate the response effort, investigate the incident, and implement containment and remediation measures.

```
root@bt:/mnt/sdb1# kwrite alert
```

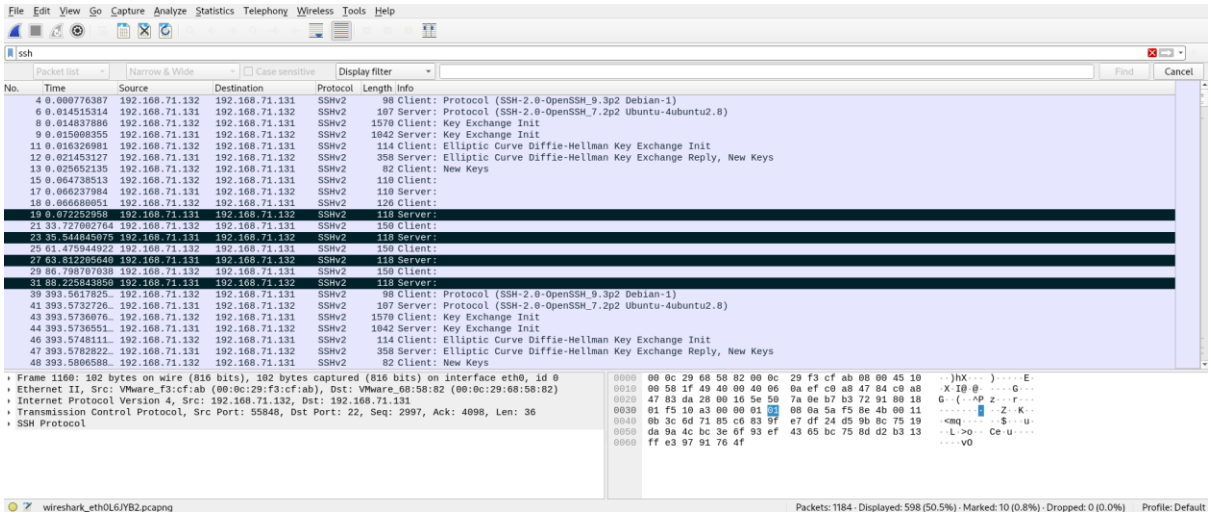
alert - KWrite

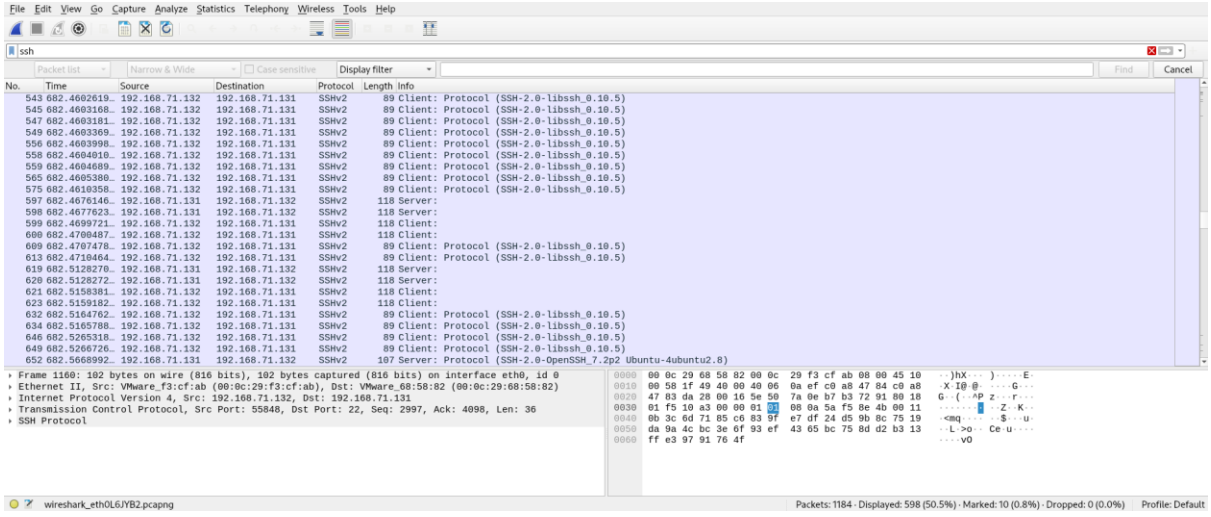
File Edit View Bookmarks Tools Settings Help

```
[**] [1:2002383:11] ET SCAN Potential FTP Brute-Force attempt [**]
[Classification: Unsuccessful User Privilege Gain] [Priority: 1].
06/08-20:28:15.796400 192.168.1.100:21 -> 216.5.1.200:1057
TCP TTL:128 TOS:0x0 ID:1995 IpLen:20 DgmLen:79 DF
***AP*** Seq: 0x6B20E2DF  Ack: 0x4681FEA6  Win: 0x100  TcpLen: 20
[Xref =>
http://www.emergingthreats.net/cgi-bin/cvsweb.cgi/sigs/SCAN/SCAN_FTP_Brute_Force][Xre
f => http://doc.emergingthreats.net/2002383]

[**] [1:2002383:11] ET SCAN Potential FTP Brute-Force attempt [**]
[Classification: Unsuccessful User Privilege Gain] [Priority: 1].
06/08-20:28:15.813277 192.168.1.100:21 -> 216.5.1.200:1057
TCP TTL:128 TOS:0x0 ID:2005 IpLen:20 DgmLen:79 DF
***AP*** Seq: 0x6B20E474  Ack: 0x4681FF53  Win: 0xFF  TcpLen: 20
[Xref =>
http://www.emergingthreats.net/cgi-bin/cvsweb.cgi/sigs/SCAN/SCAN_FTP_Brute_Force][Xre
f => http://doc.emergingthreats.net/2002383]

[**] [1:2002383:11] ET SCAN Potential FTP Brute-Force attempt [**]
```





2.0 Incident Response Template:

Date of Report: 03/09/2023

CONTACT PERSON

Full Name: Peter Cross

Address: 22 Key Rd. East Melbourne

Job Title: IT Manager ABC Retail Store

Phone: 0123456789 E-Mail: pcross@abcretails.com

THE INCIDENT

Date of Incident: 02/09/2023 Time: 8:00☒ AM ☐ PM

Type of Incident: ☐ Malware ☒ Data Breach ☐ Other: [OTHER]

How was the incident detected / discovered?
The IDS with ET SCAN signatures detected a potential brute-force attack on the SSH and HTTP services.

NOTIFICATION

Were other personnel notified? ☒ Yes ☐ No

If yes, enter: Purple Team.

CONTAINMENT

Were any containment measures made? ☒ Yes ☐ No

If yes, describe: Isolate compromised systems from the network.

IMPACTED SERVICES

Was anything permanently impacted by the incident? ☐ Yes ☒ No
No downtime or disruption to normal operations.

ATTACK VECTOR

Do you know how the attack was made? ☒ Yes ☐ No

If yes, describe: The Red Team conducted a controlled brute-force attack on the SSH and HTTP services by using the Hydra tool.

INFORMATION IMPACT

Was there any data, records, or information breached? ☐ Yes ☒ No

The incident was part of a simulated, deliberated, and controlled Red Team exercise. No actual data, records, or information were breached during this exercise.

OTHER

Is there any other information you would like to include in this report? ☐ Yes ☒ No

If yes, describe: [DESCRIBE]

3.0 System identifications:

- **Server Name/ID**: http://www.abc.com
- **IP Address:** 192.168.40.135
- **Operating System:** Linux Ubuntu 4.4.0-21-generic
- **Affected Services:** SSH (Secure Shell) – Port: 22 – Software: OpenSSH 7.2p2
  HTTP (Hypertext Transfer Protocol) – Port: 80 – Server: Apache/2.4.18
- **Vulnerabilities Exploited:**
  - Weak Passwords: the affected system permitted the use of weak and easily guessable passwords, significantly increasing the risk of unauthorised access.

4.0 Impact of security incident:

Technical Impact

Unauthorised Access:

- The Red Team gained initial access to the SSH service by exploiting weak passwords. Once inside the system, they escalated privileges, ultimately achieving root access.
- Additionally, the Red Team obtained access to the HTTP service using compromised credentials.

Potential risks:

- As result of this test, there is a significant risk of a potential data breach and unauthorised access to sensitive information. This includes employee and customer credentials, emails, and financial data.

No Downtime or Disruptions:

- It is important to emphasize that throughout the course of this Red Team exercise, there were no reported instances of system downtime or disruption to normal operations.

5.0 Actions taken for containment, recovery, and restoration of services:

Containment:

- Isolate and disconnect all Compromised Systems from the network right away to stop any further lateral movement or illegal access.
- Change all user account passwords, particularly those with privileged access to the data.
- Review and lock down access control by modifying firewall rules, access control lists, and other security configurations to limit access to only required and authorized users.
- Recognize and patch any security flaws or vulnerabilities within the network or system that may have been exploited to obtain unauthorized access to data.
- If necessary, temporarily disconnect internet connections or shut down the system and other external access points to stop further illegal access from outside sources.

Recovery

- Conducting a thorough investigation for understanding the scale of the breach, the techniques employed by the Red Team, and the amount of unauthorized access to the data.
- Rebuild or repair the hacked systems by using known secure backups and make sure that the restored systems have all the latest security patches and upgrades installed.
- To detect and respond to such attacks before affecting your system in the future, we require to strengthen security measures by putting intrusion detection systems (IDS) in place, monitoring networks, and doing better log analysis.
- For all users and employees, reinforce security awareness and training to stave against future social engineering scams.

Restoration of Services

- Reconnect systems to the network gradually after thoroughly checking and confirming their integrity. Regularly monitor for any indications of unethical behaviours.
- Conduct an audit of the whole network infrastructure and systems to make sure they are setup securely and are up to date.
- Strengthen access controls by applying two-factor authentication (2FA) and strong password guidelines.
- Document the incident, including the steps and measures taken and the lessons learned, to help you respond to incidents better in the future.

6.0 Communication with stakeholders and relevant authorities:

Internal Communications

- Make sure to immediately inform the internal security team or the IT department in charge of incident response and ensure that they are informed of the situation and capable of responding appropriately.
- Notify senior management and leadership within company, including the CEO and, if necessary, the CISO (Chief Information Security Officer) or an equivalent. Provide them a high-level outline of the circumstance.
- Share pertinent technical information with the IT and technical teams handling incident containment and recovery and collaborate with them closely throughout the process.

External communication

- Contact with stakeholders and reassure them of the measures, such as security upgrades and improved monitoring, are being done to prevent a similar situation in the future.
- Inform vendors, partners, or third-party service providers who may be affected by the incident, if relevant.
- The lack of downtime or disruptions should be highlighted as a positive aspect of the incident, demonstrating company's capacity to respond to security concerns without creating significant disruptions.
- Notify customers who have been affected, especially if their data or credentials have been compromised. Be open and honest about the issue and provide guidance on what measure they need to take.

7.0 Lessons learned and recommendations for future incidents:

- Conducting a post-event review once the incident has been solved entirely to identify lessons learned and opportunities for improvement in our security posture and incident response protocols.
- Regularly monitoring the network and system to be prepared of the situation for any new developments or threats and be ready to communicate quickly if things change.
- As cybersecurity is an ongoing process, and that the organization's defences must constantly adapt to the changing threat landscape. To remain ahead of potential threats, we need regularly review and upgrade security measures on a modern basis.
- To maintain all systems and software up to date, we require to implement a solid security patch management procedure.
- Internal and external employee awareness and education deployment on regular basis.



PURPLE TEAM Analysis

1.0 Introduction

JNML.CO conducted a comprehensive review on the ABC Retails Store's domain utilising The Red Team to exploit vulnerabilities while The Blue Team initiated response during the period of 2nd September 2023 to 16th September 2023.

The Purple Team collaborated with The Red Team and The Blue Team to facilitate an organised approach to simulating a cyber-attack on ABC Retail Store's domain. The Purple team reviewed all weaknesses and vulnerabilities discovered by The Red Team combined with the discovery tools utilised by The Blue Team to defend the network, in order to understand the ABC Retail Store's current networking landscape. This enabled The Purple Team to build a robust strategic plan to strengthen and enhance the organisations security hygiene and posture, providing increased resilience in responding to malicious attacks.

2.0 Background

The Red Team engagement utilised tactics to target the systems being assessed. This methodology encompassed a series of steps, starting with gathering open-source intelligence (OSINT), followed by system enumeration, exploitation, and executing attacks to achieve specific operational objectives. They identified multiple weaknesses and vulnerabilities in the ABC Retail Store's network including. Aligning with the OWASP Top:102021 below were exhibited in this case:

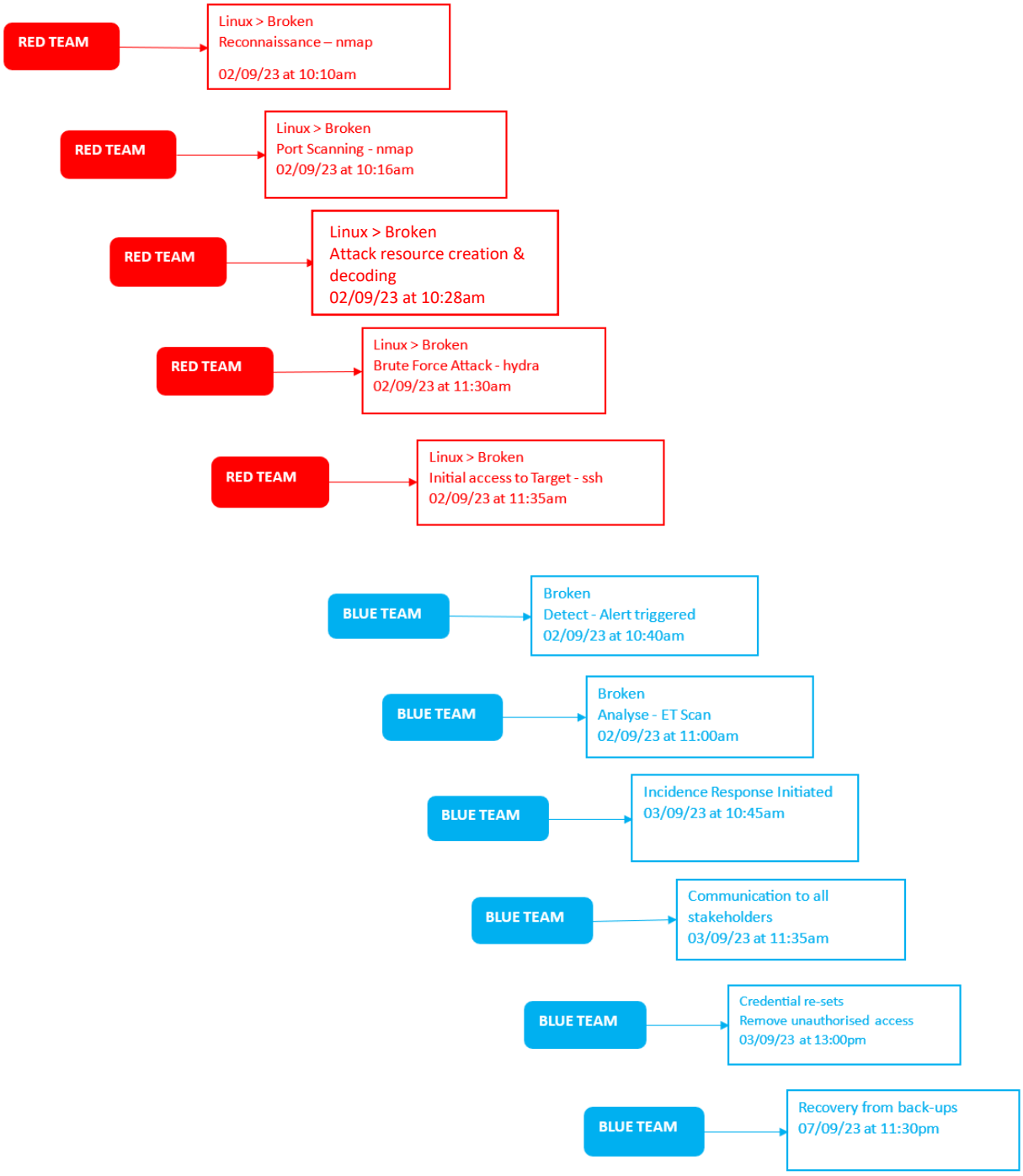
A01:2021 – Broken Access Control
Robust Access Control measures and policies ensure users act within their permissions. Failure to adopt Access Control policies leads to violations of privileges, unauthorised edit and write access etc.

A06:2021 – Vulnerable and Outdated Components
A network infrastructure is vulnerable if their physical and/or logical components are out of date or unsupported. This opens up the possibility of threats and attacks. A robust update and patching process with regular scheduled patches reduce these vulnerabilities.

A07:2021 – Identification and Authentication
It is critical to ensure user's confirmation of identity, authentication, and session management in order to protect against unauthorised attacks. Weaknesses in this area may result in brute force attacks as per CWE category: Credential Management Errors (ID 521 Weak password requirements, 256 plaintext storage of a password, 1392 Use of default credentials)
CWE - CWE-255: Credentials Management Errors (4.12) (mitre.org)

Reference:
OWASP Top 10:2021

3.0 Timeline



4.0 Red Team review

JNML.CO's Purple Team reviewed The Red Team's actions where they utilised Nmap and Hydra tool to target the ABC Retail Store's network locating multiple weaknesses and vulnerabilities. Findings included open ports, weak credentials including logins and passwords, weak password policy, lack of MFA, outdated software, lack of firewall strategy/policy, weak ACL's, weak IDS. The Red Team was able to utilise these weaknesses and vulnerabilities to gain access to The ABC Retail Store's network and perform read, write, edit actions on sensitive data and files accessed, as well as exfiltrate sensitive data from the organisations network.

Table of actions conducted by The Red Team below:

Action No.	**Date and time**	**Team**	**Host OS**	**Target OS**	**Attack method**	**Action / command**	**Category** (Attack, Detect, Prevent, Contain, Eradicate, Recover, Communicate)	**Additional Comments**
1	02/09/2023 10:10am	RED	Linux	Broken	nmap	Nmap -A -T4 -p- 192.168.40.135	Attack	Initial Reconnaissance
2	02/09/2023 10:16am	RED	Linux	Broken	nmap	Nmap -p 22 -A 192.168.40.135	Attack	Port Scanning
3	02/09/2023 10:28am	RED	Linux		N/A	cat > user.txt cat > pass.txt Python Code	Attack	Attack resource creation (list text file & Python code for automation)
4	02/09/2023	RED	Linux	Broken	N/A	Reconnaissance on HTTP 192.168.40.135	Attack	Accessed the website hosted at the IP address
5	02/09/2023	RED	Linux	Broken	xxd	xxd -r -p README.md > README.txt	Attack	File conversion and sensitive information discovery
6	02/09/2023 11:30am	RED	Linux	Broken	hydra	Hydra -L user.txt -P pass.txt 192.168.40.135 ssh	Attack	Brute Force Attack
7	02/09/2023 11:35am	RED	Linux	Broken	ssh	ssh broken@192.168.40.135	Attack	Initial network access to target

5.0 Blue Team review

The Blue Team's security monitoring systems responded to the attacks of The Red Team utilising the current IDS and Wireshark log analysis, which identified abnormal login attempt activity. In response, The Blue Team took immediate steps to contain the incident by removing access and isolating the affected network to reduce likelihood of sensitive data loss. Compromised credentials were changed immediately, and all employees were prompted for effective immediate credential updates. All ACL's were reviewed and updated security requirements, software patch updates conducted, firewall rules updated and data back-up and restore processes implemented. The Blue Team identified areas of the organisation's security posture and Cyber Hygiene that required immediate attention and improvement.

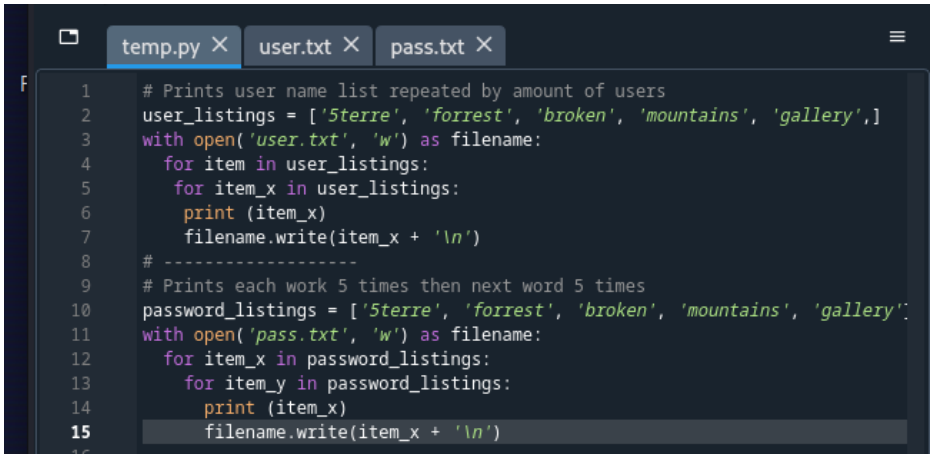
Table of actions conducted by The Blue Team below:

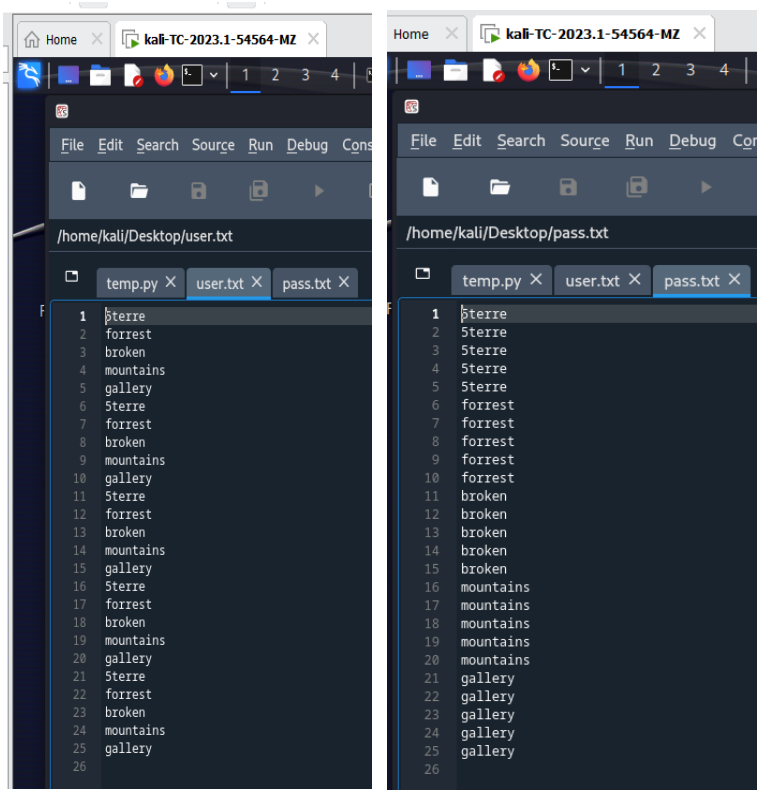
Action No.	**Date and time**	**Team**	**Target OS**	**Detect Tool**	**Action / command**	**Category** (Attack, Detect, Prevent, Contain,

						Eradicate, Recover, Communicate)
1	02/09/2023 10:40am	BLUE	Broken	IDS	Allert triggered - KWrite	Detect
2	02/09/2023 11:00am	BLUE	Broken	Wireshark	ET Scan – Brute-Force attempt	Analyse
3	03/09/2023 10:45am	BLUE	N/A		Incidence Response Initiated	Contain
4	03/09/2023 11:35am	BLUE	Broken		Communication to all stakeholders	Communicate
5	03/09/2023 13:00pm	BLUE	Broken		Credential re-sets Remove unauthorised access	Eradicate
6	07/09/2023 11:30am	BLUE	Broken		Recovery from back-ups	Recover / Restore

6.0 Purple Team exploits automation (optional):

The Purple Team have programmed a python script that automates password files for hydra tool within the use of The Red Team attack. This assists in accelerating the testing process for future assessments. Below shows steps for automation:





7.0 Purple Team actions taken

Based on the findings and reports of The Red Team and The Blue Team, The Purple Team took the following steps to enhance the organization's risk control measures:

1. Strong password policies: The Purple Team updated its password policy company-wide to prevent unauthorized access. Policies include requiring complex passwords, mandatory password updates every six months, no reuse of old passwords, and limits on the number of login attempts.

2. Two-factor authentication (2FA): The Purple team implemented two-factor authentication system to provide additional security against unauthorized access.

3. Regular Security Updates: The Purple Team has initiated a program to regularly update all software, hardware, and security patches of the entire network system within the organisation. Update and patches will be regularly scheduled to ensure most up to date versions implemented.

4. Encryption: The Purple Team encrypted all sensitive data, both transmitted and at rest. Additionally, removed unsecured data transfer paths.

5. Access Control: The Purple Team has strengthened access controls. Robust access controls based on roles was added, removing unnecessary and default access based on least privilege principle.

6. Monitoring and Logging: The Purple Team upgraded the network monitoring system and IDS (Intrusion Detection System) to continuously monitor system activity and network traffic. Firewalls were reconfigured to improve detection and respond to unauthorized access.

7. Incident Response Plan: The Purple Team reviewed and renewed the incident response plan to help organizations respond faster and more efficiently to cyber-attacks.

8. Employee Training: The Purple Team provided employees with a regular training program to increase employee awareness of password security. Through training, employees are educated on best practices for protecting sensitive data.

8.0 Recommended presentation

1. Disable unnecessary or unsecured access paths: For example, disable HTTP and use HTTPS for more secure connections. It will mitigate the potential security risks associated with unsecured connection.
2. Access control configuration: use Two/multiple factor authentication to ensure that the access is legal. It's hard for hackers to obtain additional information about the authentication.
3. Continuous monitoring: organisation should implement more secure firewall settings to block the abnormal traffic flow into the internal network.
4. Network Security: network can be separated into multiple zones and using firewalls to control the traffic in and out of the different zones. It will significantly reduce the malicious traffic from the outside network.
5. Least Privilege Principle: only authorise the least privilege to the person so that even the hackers steal one account credential, they cannot get all sensitive information from the network.
6. Regular Audits and Penetration Testing: regular audits and penetration testing will help organisation renew its security policies and controls to meet the needs of the security.
7. Security Awareness and Training: employee education is important for improving security posture in the organisation. Employees should be trained to understand the importance of the password security.
8. Regular backup: critical data should be backup regularly and backup should be stored in the safe environment. Even if an organization's data is damaged, it can be quickly restored.

9.0 Conclusion

The Purple Team coordinated The Red and The Blue Teams to achieve efficient collaboration between offensive and defensive personnel. Based on the review of the work of The Red and The Blue Teams, The Purple Team successfully enhanced the ABC Retail Store's organizational network security posture by implementing enhanced password policies, cyber hygiene, access control, two-factor authentication, firewall configuration, scheduled security updates and patches, encrypted data, employee training, and conducted continuous monitoring.

Cybersecurity is an ongoing endeavour due to the constantly evolving cyber threat landscape. In response to evolving threats, The Purple Team provides organizations with security recommendations, including but not limited to conducting regular reviews of cybersecurity policies and taking appropriate measures to address different security risks.

In short, this offensive and defensive exercise demonstrated the organization and coordination role of the purple team. Through a series of initiatives taken by The Purple Team, the organization's

security posture has been strengthened and can respond to changing security threats more efficiently. This exercise will be utilised to form security strategies against future cyber threats for ABC Retail Store.

Signed: [Team Leader Name]



[Team Leader Signature]



Date: [Date of Report]

11/10/2023